Roll No. | | | | | | | |

END SEMESTER EXAMINATION

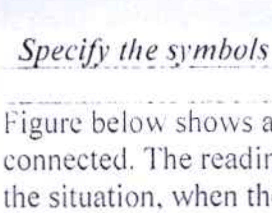
Name of the Course: B.Tech Aerospace Engineering | *Semester: III*

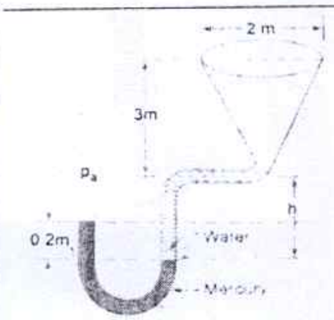
Name of the Paper: Aero Fluid Mechanics | *Course Code:* TAS-304

Time: Three Hours | MM: 100

Note:

i. All questions are compulsory.
ii. Answer any two sub questions in each main question.
iii. Total marks for each main question is **twenty**
iv. Each Question carries **10 marks**

Q1		COs
a.	Explain the following with their SI Unit *i.* *Viscosity* *ii.* *Kinematic Viscosity* *iii.* *Specific Weight* *iv.* *Specific gravity* v. *Bernoulli's equation*	CO1, CO2
b.	Calculate the dynamic viscosity of oil, which is used for lubrication between a square plate of side 0.8 m and an inclined plane with an angle of inclination 30° as shown in the figure below. The weight of the square plate is 300 N and it slide down the inclined plate with a uniform velocity of 0.1 m/s. The thickness of the oil film is 1.5 mm.  	CO1, CO2
c.	Explain the hydrostatic law? Derive an expression for difference pressure for the inverted U-tube differential manometer. *Specify the symbols used during derivation*	CO3
Q2		
a.	Figure below shows a conical vessel having to which a U-tube manometer is connected. The reading of the manometer indicated in the figure pertains to the situation, when the vessel is empty. Find the reading of the manometer when the vessel is completely filled with the water. What will be the reading if it is completely filled with a crude oil of Sp. Gr = 0.9 in place of water.	CO3, CO4

		
b.	In a fluid. the velocities field is given by $V = (3x + 2y)i + (2z + 3x^2)j + (2t - 3z)k$ Determine: i. The velocity component u. v. w at any point in the flow field ii. The speed at point (2.1.1) iii. The speed at time t = 4s at point (0.0.2)	CO1, CO4
c.	Find the discharge through a trapezoidal notch which is 1 m wide at the top and 0.4 m at the bottom and id 30 cm in height. The head of the water on the notch is 20 cm. ***Assume Cd = 0.62 and 0.6 for the rectangular and triangular portion respectively***	CO1, CO5
Q3		
a.	Derive an expression for the ***Bernoulli's equation*** by using the ***Euler's equation*** of motion of fluid element. Explain with the diagram *Specify the symbols used during derivation*	CO1
b.	A horizontal venturimeter with inlet and throat diameter 30 cm and 15 cm respectively is used to measure the flow of water. The reading of differential manometer connected to the inlet and the throat is 20 cm of mercury. Determine the rate of flow (discharge) through the pipe	CO1, CO2
c.	Define fluid kinematics? What are the descriptions of different fluid motion? Illustrate Langrangian methods of fluid motion in kinematic flow of fluid?	CO2, CO5
Q4		
a.	What do you mean by viscous fluid of flow? *Derive an expression for the* Pressure drop over a given length of pipe (L) i.e., ***the equation of Hagen Poiseuille***	CO1, CO5
b.	An oil of viscosity $0.1 Ns/m^2$ and relative density of 0.9 is flowing through a circular pipe of diameter 50 mm and length 300 m. The rate of fluid through a pipe is 3.5 liter/s. Find the pressure drop in a length of 300 m and also the shear stress a the pipe wall?	CO4, CO5
c.	Discuss the classifications of loss of energy in pipes? Find the head lost due to friction in a pipe of diameter 300 mm and length 50 m through which water is flowing with a velocity of 3 m/s by using *Darcy formula*	CO1, CO4
Q5		
a.	Define the following: a. Boundary layer thickness	CO4, CO5

	b. Displacement thickness c. Momentum thickness d. Energy thickness	
b.	Explain ***Von Karman momentum Integral equation***, and its application? Derive the same equation with suitable diagram?	CO4, CO5
c.	What do you mean by separation of boundary layer? Discuss effect of pressure gradient on boundary layer separation with neat and clean diagram?	CO2, CO5